KAZI RAKIB UDDIN

M.S. Candidate, Aquaculture and Aquatic Sciences, Kentucky State University
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RESEARCH INTERESTS

Fish physiology and gene expression under environmental stress; salinity and water quality in recirculating systems; fish and shrimp nutrition; fish health and histology; fish reproduction and hatchery production.

EDUCATION

M.S., Aquaculture and Aquatic Sciences Expected May 2027
Kentucky State University, Frankfort, KY | GPA: 4.00/4.00
Thesis: Effects of salt type and salinity on performance and molecular response in olive flounder (*Paralichthys olivaceus*). Advisor: Dr. Andrew J. Ray

M.S., Aquaculture Jul 2024
Sylhet Agricultural University, Bangladesh | GPA: 3.81/4.00
Thesis: Dietary inclusion effects of fermented major oil seed byproducts on growth and health performance of climbing perch *Anabas testudineus*. Advisor: Dr. Md. Sakhawat Hossain

B.Sc., Fisheries Nov 2022
Sylhet Agricultural University, Bangladesh | GPA: 3.62/4.00

RESEARCH EXPERIENCE

Graduate Research Assistant, Kentucky State University Jan 2025 – Present
USDA Evans-Allen project: practical methods for enhancing productivity, sustainability, and profitability of marine aquaculture systems to grow healthy food

- Run a multi-treatment salt type and salinity trial on olive flounder (*Paralichthys olivaceus*) in 16 recirculating tanks, with sampling every two weeks.
- Monitor water quality with YSI meters and nitrogen tests (ammonia, nitrite, nitrate); track ammonia thresholds across pH, temperature and salinity and adjust system operations.
- Grow microalgae, *Ulva* and *Salicornia* on aquaculture sludge and measure nitrogen and phosphorus recovery.
- Perform gill and liver histology, RNA extraction, cDNA synthesis and qPCR gene expression analysis; analyze data in R.
- Raise Pacific white shrimp (*Litopenaeus vannamei*) at high stocking density in a 20 m^3 intensive saltwater raceway; build and maintain RAS.

Research Assistant, Sylhet Agricultural University Jan 2023 – Jul 2024

- Formulated diets with fermented mustard oil cake and fermented soybean meal as alternative protein sources for climbing perch (*Anabas testudineus*); ran the feeding trial and analyzed proximate composition, growth, hematology, immune parameters and histology.

Research Assistant, Sylhet Agricultural University Mar 2022 – Dec 2022

- Evaluated black soldier fly larvae oil as an alternative lipid source for butter catfish (*Ompok pabda*); measured growth and health responses.

HATCHERY AND FIELD EXPERIENCE

- Spawned muskellunge (*Esox masquinongy*), sauger (*Sander canadensis*) and koi, and cared for koi eggs and fry, in the Fish Reproduction and Spawning Techniques course, Kentucky State University (Spring 2026).
- Managed broodstock, harvested fingerlings, seined, operated aeration equipment, and transported live fish by hauling truck to different locations.
- Electrofished the Kentucky River through graduate coursework; helped identify native and invasive fishes, including invasive carp, from tailwater collections at the North American Native Fishes Association Convention (June 2025).

PUBLICATIONS

Helen ST, Dey T, Bharoteshwari A, **Uddin KR**, Kabir MA, Hasan MR, Hossain MS. 2026. Evaluation of different terrestrial oils as an alternative to dietary fish oil on feed physical properties, growth, feed utilization, and fatty acid profile of Gangetic catfish (*Mystus cavasius*). *Animals*, 16(2), 330.

Hridoy MAA, Paul PB, Rasel MA, **Uddin KR**. 2025. Impact of water pollution on physio-chemical properties for fish habitat in the Surma River, Sylhet City, Bangladesh. *Water and Environmental Sustainability*, 5(1), 12–20. doi:10.52293/WES.2025.1.1042

CONFERENCE PROCEEDINGS

Uddin KR*, Dey T, Hossain MS. "Performance evaluation of fermented mustard oil cake and fermented soybean meal as alternative protein sources in climbing perch (*Anabas testudineus*)". World Aquaculture Society, Aquaculture America 2025, New Orleans, Louisiana, USA. March 2025. (Oral)

Bharoteshwari A, Dey T, Hossain MS, **Uddin KR***. "Dietary inclusion effect of black solder fly larvae (bsfl) oil as an alternative lipid source on growth and health performance of butter catfish (*Ompok pabda*)". 13th Indian Fisheries and Aquaculture Forum, Kolkata, India. February 2024. (Oral)

Uddin KR*, Kawser MA. "Increasing fish disease burden and use of antibiotics in pond-based aquaculture in South Chattogram, Bangladesh". 2nd International Conference on Sustainable Fisheries, Sylhet, Bangladesh. September 2022. (Poster)

TECHNICAL SKILLS

- **Aquaculture:** recirculating system operation, intensive shrimp raceway, hatchery and broodstock management, spawning, seining, fish transport
- **Water chemistry:** YSI multiparameter monitoring; ammonia, nitrite, nitrate, alkalinity, dissolved oxygen, salinity; nitrogen compounds
- **Molecular:** DNA/RNA extraction, cDNA synthesis, qPCR gene expression analysis, dPCR, Illumina library preparation; hands-on Oxford Nanopore and Illumina sequencing (graduate coursework)
- **Fish health:** hematology, immune parameters, tissue histology
- **Nutrition:** feed formulation and preparation, alternative ingredients, feeding trials, proximate composition
- **Data:** R, SPSS, STATA, Microsoft Excel; experimental design, ANOVA and regression; data entry and quality control

RELEVANT COURSEWORK

Kentucky State University (graduate): Fish Reproduction and Spawning Techniques; Molecular Techniques for Environmental and Aquatic Studies; Fish Morphology and Physiology; Recirculating Aquaculture; Water Quality Management; Fish Nutrition; Applied Statistics.

Sylhet Agricultural University: Fish Breeding and Genetics; Principles of Genetics; Fish Hatchery Management; Fish Physiology; Freshwater Aquaculture; Biochemistry; Aquaculture Nutrition (graduate).

TEACHING

Teaching Assistant, Fish Nutrition Lab, Sylhet Agricultural University Jul 2023 – Jun 2024

- Taught undergraduates proximate composition methods and instrumentation; ran and graded lab exams.

AWARDS

- National Science and Technology (NST) Fellowship, Ministry of Science and Technology, Bangladesh, 2023–2024 (M.S. thesis research grant)

TRAINING, OUTREACH AND LEADERSHIP

- Vice President, U.S. Aquaculture Society Student Subunit, Kentucky State University (2025 – present): organized the 2026 KSU Shrimp Farming Symposium and student career events with Kentucky Department of Fish and Wildlife Resources personnel

REFERENCES

Dr. Andrew J. Ray, Associate Professor and Chair, School of Aquaculture and Aquatic Sciences, Kentucky State University. andrew.ray@kysu.edu, (502) 597-8109

Dr. Waldemar Rossi Jr., Research Professor, College of Agriculture, Health, and Natural Resources, Kentucky State University. waldemar.rossi@kysu.edu, (502) 597-8130

Dr. Md. Sakhawat Hossain, Professor, Department of Aquaculture, Sylhet Agricultural University. sakhawat.aqua@sau.ac.bd, +880 1774 949483